ZUKUNFT LAND

North Hessian Loess Plain

Full data report

Generated 2026-11-04

Hesse

NUTS-3 regions: DE734, DE737



Figure 1: The North Hessian Loess Plain Living Lab: study region (left) and its location within Germany (right). Basemap: Tiles © Esri - Source: Esri, i-cubed, USDA, USGS, AEX, GeoEye, Getmapping, Aerogrid, IGN, IGP, UPR-EGP, and the GIS User Community

Agriculture

Key figures

CROPLAND AREA	NUMBER OF FARMS	AVERAGE FARM SIZE	SHARE OF ORGANIC FARMING
62,680 ha	1,493 count	59.4 ha	12.7 %

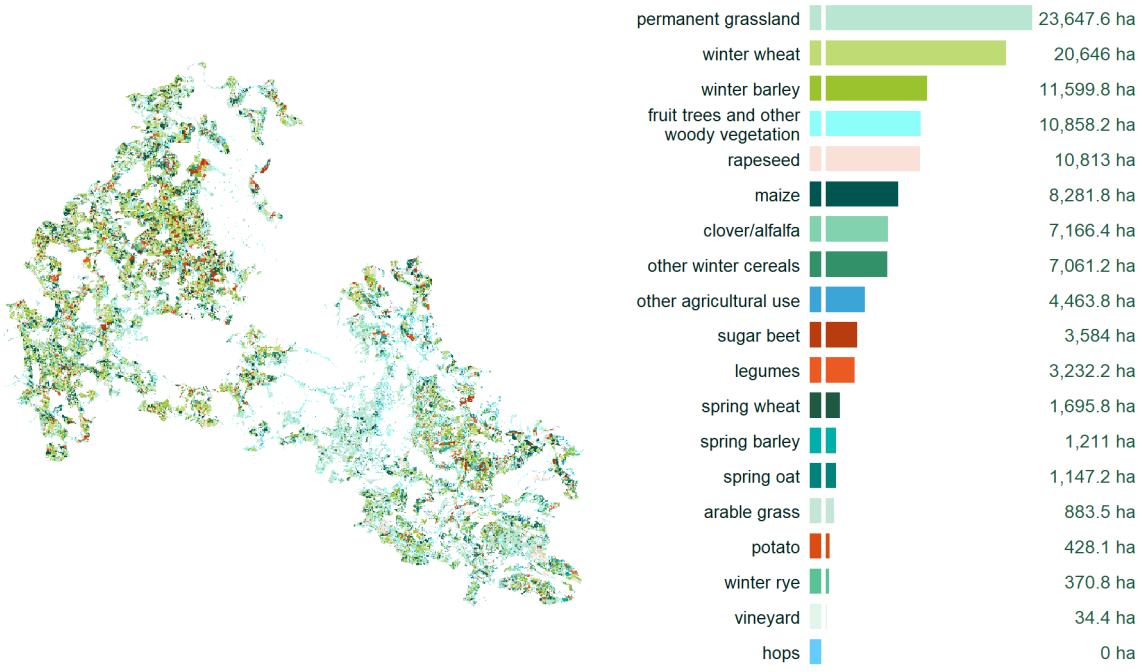


Figure 2: Map showing Agriculture in the North Hessian Loess Plain Living Lab (Data: Crop types (DLR, 2024))

Overview

Farming is small-structured and cereal-dominated: winter wheat, winter barley and winter rape form the backbone of the rotation, with cereals on around two thirds of the arable area, alongside fodder cropping, oilseeds and a widening range of niche crops — the Werra-Meißner-Kreis is Hesse's leading hemp-growing district. Organic farming is unusually strong and nearly doubled between 2009 and 2019, reaching 12% of farms and 14.4% of the agricultural area in the Werra-Meißner-Kreis, well above the national level; it is anchored by the University of Kassel's Faculty of Organic Agricultural Sciences in Witzenhausen — the first of its kind worldwide — the state domain Frankenhausen and the Ökolandbau-Modellregion Nordhessen. Livestock density is comparatively low and falling, with dairy in sharp retreat while suckler cows, sheep, horses and poultry gain ground. Roughly a third of the arable land and more than half of the grassland is under agri-environment contracts or organic management.

Challenges

Profits have been below the long-run average in most recent years, so many farms have been living off their substance; full-time holdings are disappearing faster than part-time ones and succession is often unresolved. The collapse of dairy — 54% of dairy holdings and a quarter of the cows lost in ten years — strips value

creation out of the region and removes the grazing that keeps grassland open. Organic expansion has hit a demand ceiling: producer prices for several organic products are too low, so the trend could reverse unless regional outlets such as the Kassel conurbation are opened up. Narrow cereal rotations raise pesticide use and depress humus and biodiversity, while farmland continues to be lost to road and logistics construction.

Climate

Key figures

GROWING DEGREE DAYS, ANNUAL SUM (BASE 5 °C)	MEAN ANNUAL TEMPERATURE	ANNUAL PRECIPITATION	PRECIPITATION OF WARMEST QUARTER
1,735 °C·d +1,076 °C·d by 2071-2100	**8.5** °C +3.9 °C by 2071-2100	**805** mm +2.3 % by 2071-2100	**227** mm -8.7 % by 2071-2100

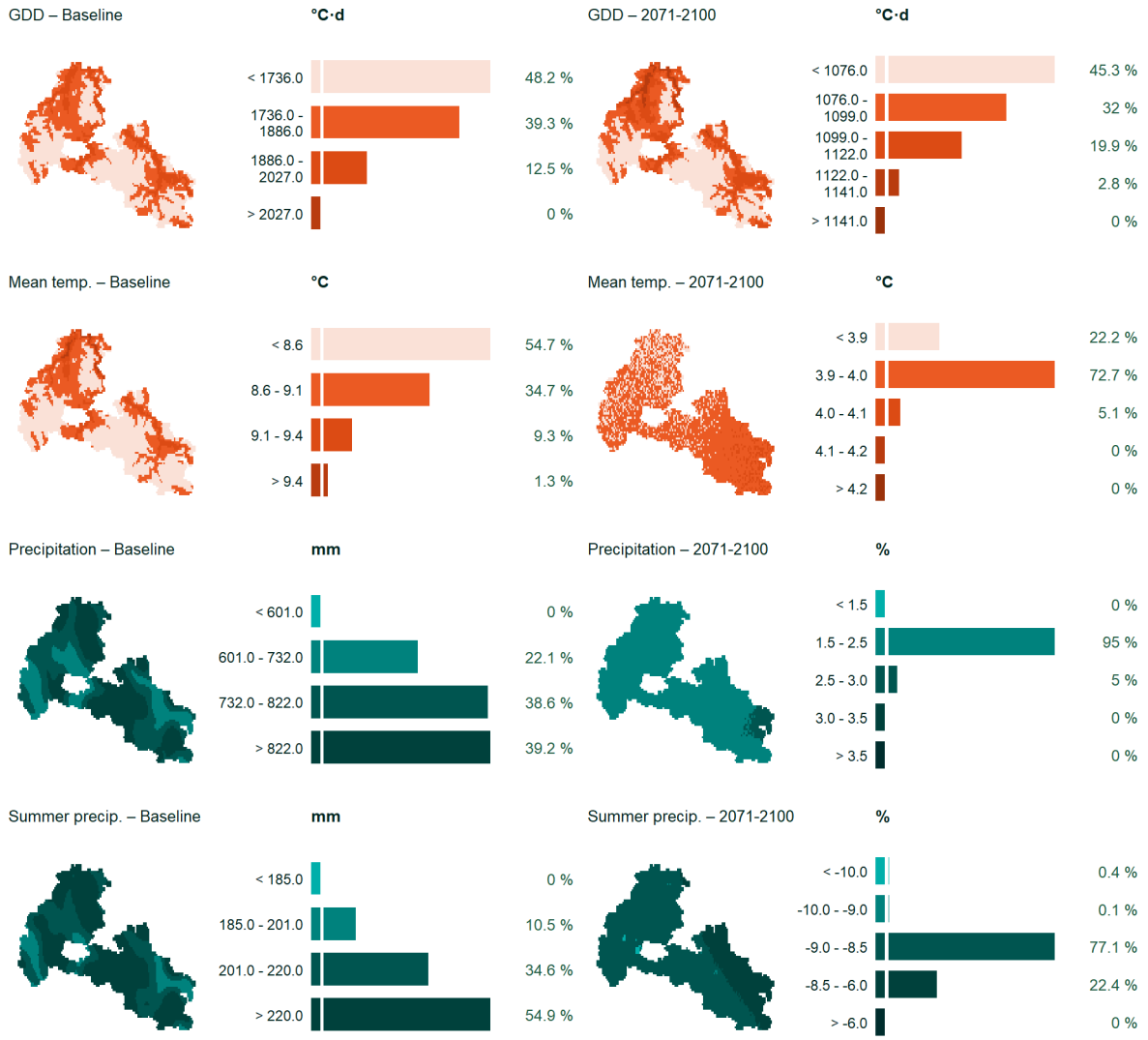


Figure 3: Map showing Climate in the North Hessian Loess Plain Living Lab (Data: Climate (CHELSA V2.1 / CMIP6): Growing degree days, Mean annual temperature, Annual precipitation, Precipitation of warmest quarter)

Heat accumulated above 5 °C over the year - a measure of how much growing season a crop gets.
Mean annual air temperature, averaged over the reference period.
Total precipitation summed over the year.
Precipitation falling in the warmest three-month quarter - relevant to summer drought stress.

North Hessian Loess Plain

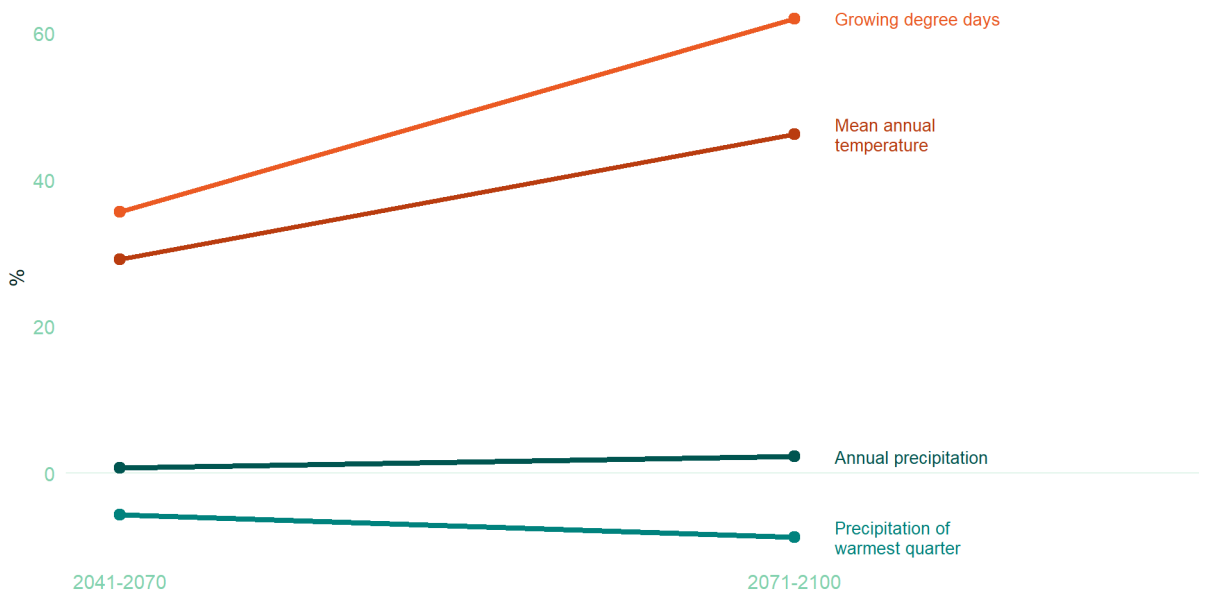


Figure 4: Chart showing Climate in the North Hessian Loess Plain Living Lab (Data: Climate (CHELSA V2.1 / CMIP6): Growing degree days, Mean annual temperature, Annual precipitation, Precipitation of warmest quarter)

Overview

The basins lie in the rain shadow of the surrounding uplands and are markedly dry: 500—600 mm of annual precipitation in the Kassel basin and the loess Börde areas, against 800—900 mm on the heights of the Habichtswald, Kaufunger Wald and Meißner. The annual mean is close to 9.5 °C in the lowland basins and distinctly cooler at altitude, where the growing season starts later. Hesse's annual mean temperature rose from 8.2 °C in 1961—1990 to 9.3 °C in 1991—2020, and precipitation is spread fairly evenly through the year with a slight summer maximum.

Challenges

Rainfall that could once be taken for granted has become incalculable; the drought of 2018 marked the turning point, and both farming and forestry now face redesigning their cropping and stand systems around it. Low precipitation in the basins combined with rising evaporation leaves little buffer, and on the shallow soils outside the loess areas dry spells cut into yields quickly. Growing yield uncertainty is itself shifting farm strategy: agri-environment payments have become a more predictable source of income than the harvest.

Soil

Key figures

NITROGEN SURPLUS	PHOSPHORUS SURPLUS	GROUNDWATER ABSTRACTION (NON-PUBLIC SUPPLY)
No data available kg N/ha UAA	**No data available** kg P/ha UAA	**4,023** 1000 m3

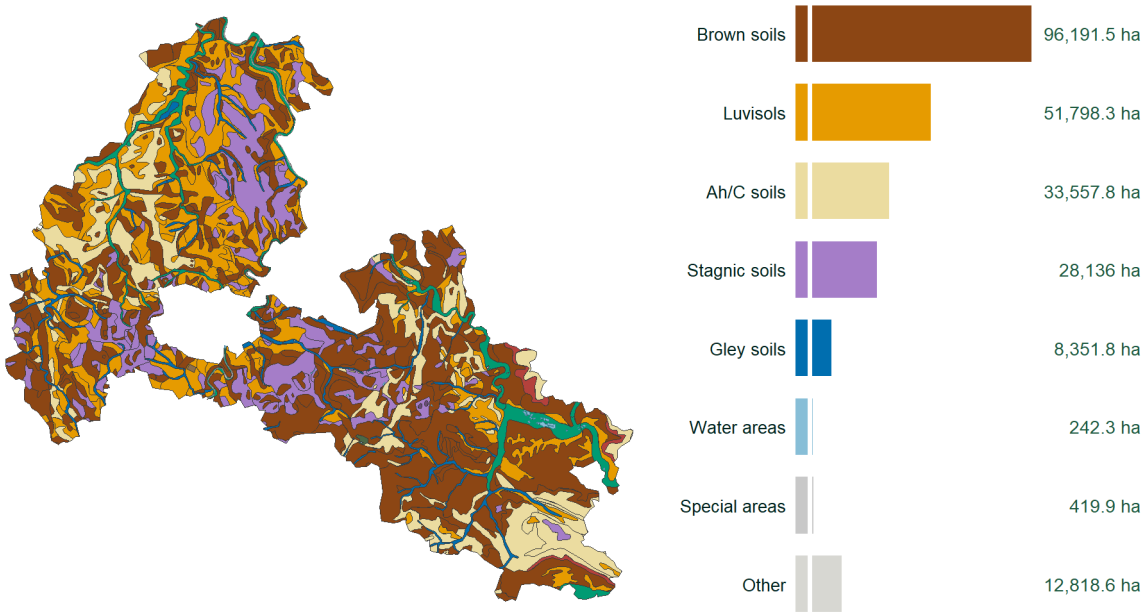


Figure 5: Map showing Soil in the North Hessian Loess Plain Living Lab (Data: Soil overview map (BUEK250))
Legend shows the dominant semantic soil groups for this Living Lab; raw BUEK IDs stay in the data only as provenance.

Overview

Loess is the region's defining resource. In the basins — the Kassel basin, the Hofgeismar depression and the Werra valley — decalcified loess has weathered into Parabraunerden with high water and nutrient storage capacity and soil ratings around 70—80 of a possible 100, among the best arable soils in Hesse. Away from the loess cover the picture changes over short distances: brown soils on Bunter sandstone are sandy, acid and largely under forest, shallow rendzinas on Muschelkalk are dry, basalt-derived soils on the Habichtswald and Meißner are stony but nutrient-rich, and alluvial loams line the Werra and Fulda valleys.

Challenges

The loess soils have been cultivated for centuries and are already truncated by erosion; on undulating terrain heavy rain moves soil off the slopes, which is why erosion-control and buffer strips are a standing element of the agri-environment programme. Outside the loess basins yield potential is low, and the shallow soils store little water, so they are the first to suffer in dry years. Cereal-dominated rotations depress humus content, while the region's low livestock density means little farmyard manure is available to rebuild it.

Socio-economic

Key figures

TOTAL POPULATION	GDP PER CAPITA	UNEMPLOYMENT RATE	DISPOSABLE HOUSEHOLD INCOME PER CAPITA
328,797 persons	37,305.5 EUR	5.05 %	26,456 EUR

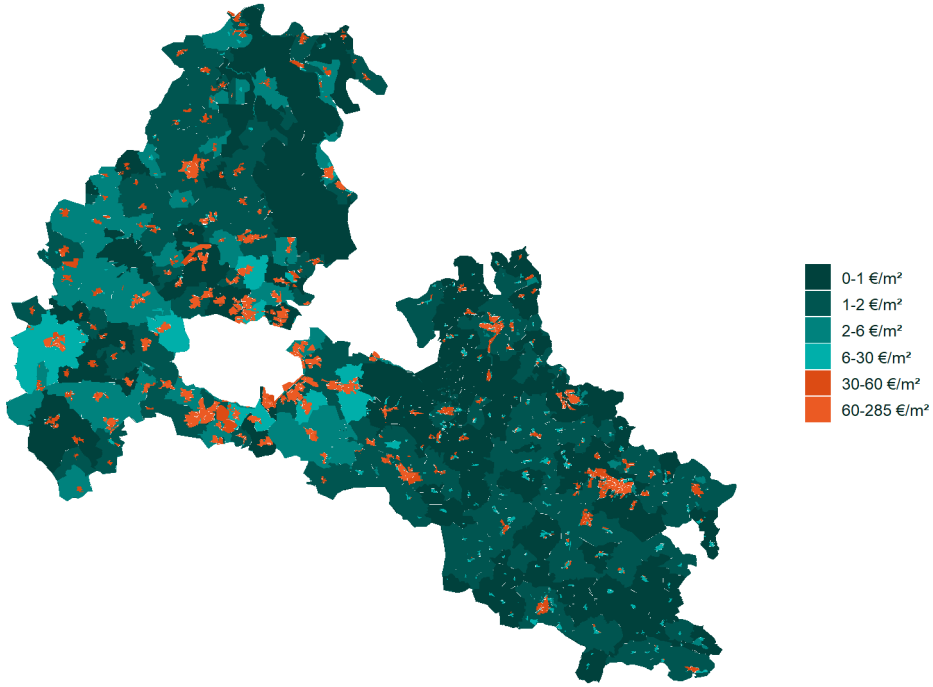


Figure 6: Map showing Socio-economic in the North Hessian Loess Plain Living Lab (Data: Standard land values (BORIS))

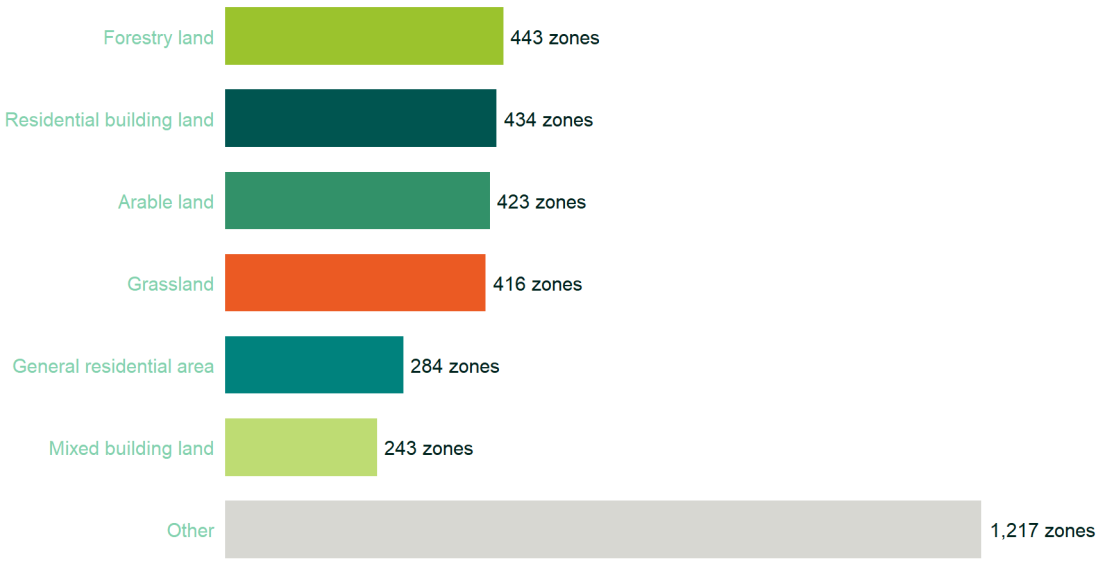


Figure 7: Chart showing Socio-economic in the North Hessian Loess Plain Living Lab (Data: Standard land values (BORIS))

Overview

Around 330,000 people live in the two districts — some 230,000 in Landkreis Kassel and 100,000 in the Werra-Meißner-Kreis — which surround, but do not include, the city of Kassel, North Hesse's only large city and its market and service centre. The Kassel area is driven by the mobility industry: the Volkswagen component plant at Baunatal is the largest employer in North Hesse with some 16,500 staff, alongside rail technology, logistics, environmental and energy technology and the University of Kassel. The Werra-Meißner-Kreis is more rural and industrially finer-grained, with automotive suppliers, tool and mould making, plastics and paper processing and refractories, plus health and wellness, logistics and organic agriculture. Tourism builds on half-timbered towns, the Reinhardswald and the Geo-Naturpark Frau-Holle-Land.

Challenges

As a former border region, the Werra-Meißner-Kreis carries long-standing structural weakness: its employment base has shrunk, the population has been declining since the 1970s, and ageing and out-migration are reinforced by the distance to larger centres. Skilled labour is scarce and farm succession frequently unclear. The food value chain is highly concentrated on the buyer side, so little processing value stays in the region; regional and organic marketing initiatives — farmers' evening markets, school catering, regional organic brands — are being built precisely to recapture it. The nearby Kassel conurbation is the obvious market for these products but is not yet fully tapped.

Landscape

Key figures

FOREST AREA	NATURA 2000 AREA	NATURE RESERVES AREA	SHARE OF SEALED SURFACE
94,673 ha	48,127 ha	6,002 ha	13.3 %

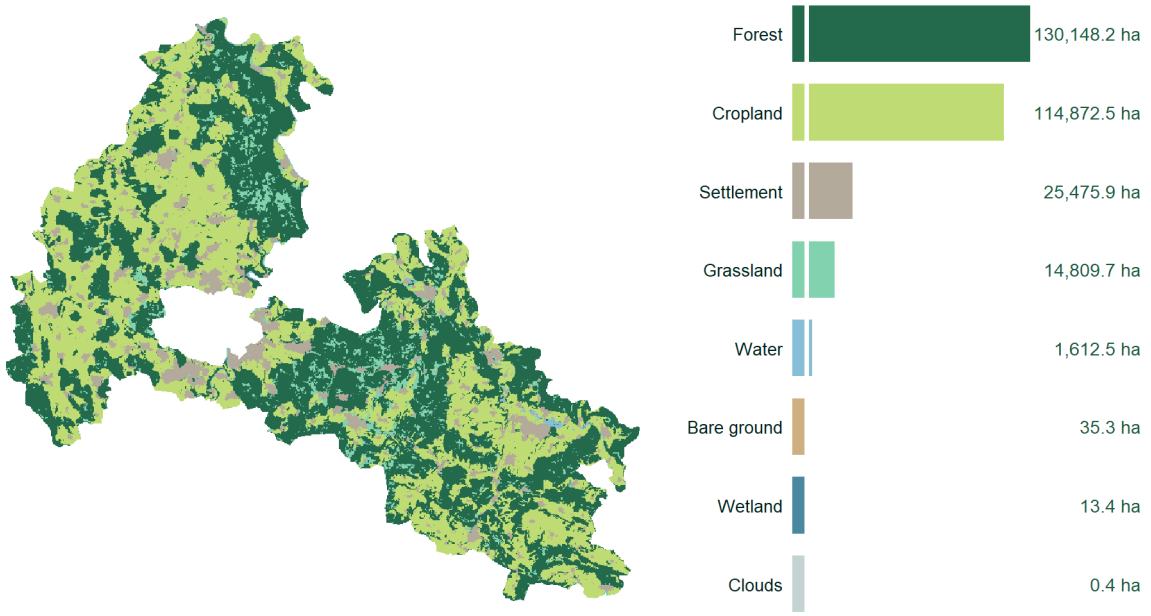


Figure 8: Map showing Landscape in the North Hessian Loess Plain Living Lab (Data: Land cover (Esri/Impact Observatory, 2024))

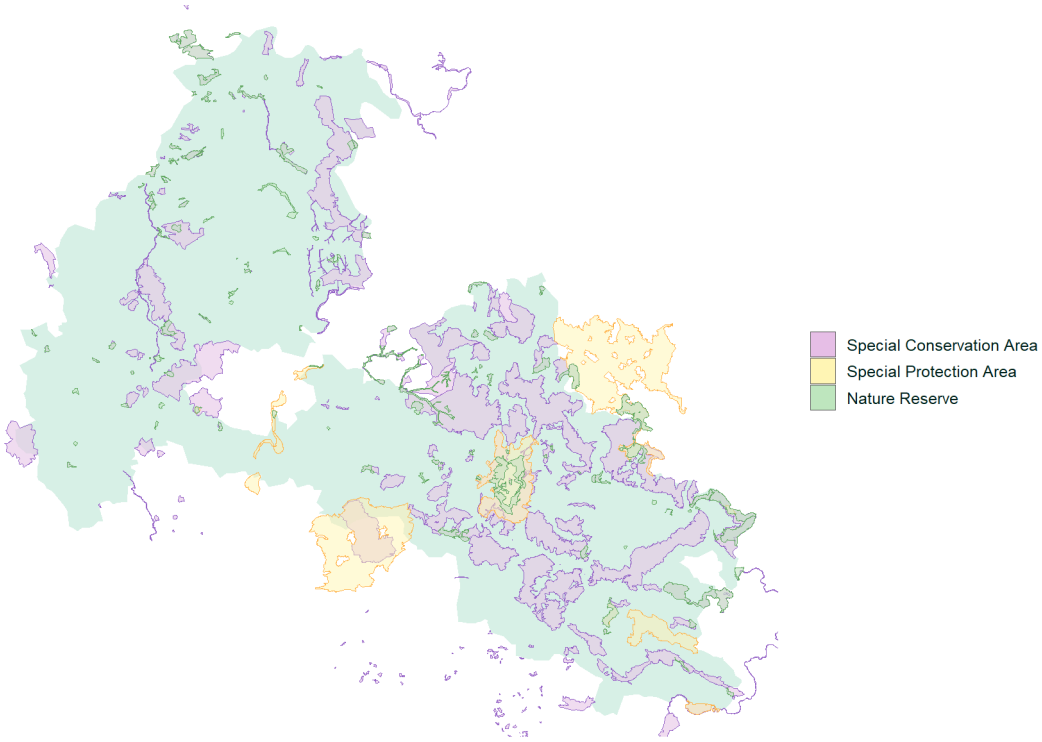


Figure 9: Map showing Protected Areas in the North Hessian Loess Plain Living Lab (Data: Protected areas (BfN, 2019/2023))

Conservation sites intersecting this Living Lab region. Full polygon boundaries are shown, so sites extend beyond the region outline and designations may overlap. Source: BfN (FFH and bird sanctuaries 2019, nature reserves 2023) - not suitable for planning purposes.

Overview

A mosaic of fertile loess basins and wooded uplands. Open, large-field farmland fills the Kassel basin, the Hofgeismar depression, the Wolfhagen basin, the Werra valley and the Lichtenau plateau, framed by the Reinhardswald, the Habichtswald, the Kaufunger Wald, the Hoher Meißner (754 m) and the Ringgau and Stölzinger uplands. Forest covers close to 40% of Landkreis Kassel, and the Werra winds through the eastern part of the region. The Naturpark Habichtswald and the Geo-Naturpark Frau-Holle-Land — with more than 200 geotopes and counting as one of Germany's 30 biodiversity hotspots — together with historic half-timbered towns give the region a strongly layered cultural landscape.

Challenges

The small-structured farmland with its wide species spectrum is one of the region's real assets, but it depends on continued grazing and extensive use that is not adequately rewarded economically; where it stops, open grassland and its semi-natural habitats are lost. Infrastructure steadily consumes farmland — motorway construction and a planned logistics site alone drove most of the roughly 560 ha of agricultural area lost in the Werra-Meißner-Kreis over ten years, and a planned high-voltage cable corridor would add years of disruption. On the arable land, narrow rotations reduce structural diversity, and the returning wolf adds uncertainty for grazing livestock keepers.